

CS-GY 9223I (CS-UY 3943)

Algorithmic Machine Learning + Data Science

Instructor: Prof. Christopher Musco (cmusco@nyu.edu)

Office Hours: 3-5pm Thursdays (except tomorrow)

Reading Group: TBD. Poll to gauge interest and time.

Webpage: chrismusco.com/9223-2019

NYU classes for forum

Exams: Midterm - 10/23, in class

Final - 12/18, class time

check for
conflicts
now!

Course Topic: Algorithmic methods for machine learning + data analysis at scale.

- High throughput / realtime data applications

(think Shazam, Google maps / waze, Amazon product recs, industrial robotics, FinTech, scientific applications)

- More complex models → more training data

(deep neural networks, reinforcement learning, machine translation)

- Data analysis on low compute devices

(smart phones / watches, robots / drones / etc., sensor networks)

Some numbers:

- Google receives $\approx 10,000$ Maps queries every second
- NASA collects 6.4 Tb of satellite images every day
 - new "ImageNet" dataset every 3 days
- Large Synoptic Survey Telescope : 15 Tb of images per night
- Broad Institute sequences 24 Tb of genetic data per day

"needle in a haystack problem"

Ushering in new golden age for research in computational methods, using entirely new tools!

Course Objectives:

1 New algorithmic toolkit (randomization, sketching, optimization, spectral methods, etc.)

- lectures - reading

See cs.princeton.edu/courses/archive/fall18/cos521

2. Learn to apply tools in the wild (industry, academia)

- 4 Problem Sets - in class work

→ midway into class

→ break into groups (auditors included)

→ pset like problem to solve

3. Theory as an approach to algorithm design.

What we won't cover:

- Software tools or frameworks
(MapReduce, Tensorflow, Amazon AWS)
(CS-64 6513: Big Data)

- Machine learning models
(neural nets, RL, Bayesian methods, unsupervised learning, function fitting, etc.)

Unit 1: The Power of Randomness

Hashing (+ Load Balancing)

- Work horse of the modern web
- good probability review!

Probability review:

X takes values in set $S \subseteq \mathbb{R}$. Eg. $S = \{1, 2, 3, 4, 5, 6\}$ for dice roll.

Expectation: $\mathbb{E}[X] = \sum_{w \in S} \Pr[X=w] \cdot w$

Continuous
r.v.s : $\mathbb{E}[X] = \int_{y \in \mathbb{R}} p(y) dy$

Independence

Two random events

A, B are

independent

if

$$\Pr(A|B) = \Pr(A)$$

✓
"A given B"

$$\Pr(A|B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

So equivalently, when A and B are independent,

$$\frac{\Pr(A \cap B)}{\Pr(B)} = \Pr(A) \rightarrow \Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$$

Roll 2 dice. What's the probability the first is odd and the second is < 3 ?

Independence Given random variables X and Y taking values in S_x and S_y . We say X and Y are independent if for all $w \in S_x, z \in S_y$ $[X=w]$ and $[Y=z]$ are independent random events.

Expectation Identities

$$\mathbb{E}[aX] = a \mathbb{E}[X]$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] ?$$

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y] ?$$

$E[X + Y]$ is true for any X, Y .

$$= \sum_{w \in S_X} \sum_{z \in S_Y} \Pr(X=w, Y=z) \cdot (w+z)$$

$$= \sum_w \sum_z \Pr(X=w, Y=z) \cdot w + \sum_w \sum_z \Pr(X=w, Y=z) \cdot z$$

$$= \sum_w w \cdot \underbrace{\sum_z \Pr(X=w, Y=z)}_{= \Pr(X=w)} + \sum_z z \cdot \underbrace{\sum_w \Pr(X=w, Y=z)}_{= \Pr(Y=z)}$$

$$= E[X] + E[Y]$$

$E[XY] = E[X]E[Y]$ true for independent r.v.'s.

$$E[XY] = \sum_{\omega \in S_X} \sum_{z \in S_Y} \Pr(X=\omega, Y=z) \omega z$$

$$= \sum_{\omega} \sum_z \Pr(X=\omega) \Pr(Y=z) \omega \cdot z$$

$$= \sum_{\omega} \left[\omega \cdot \Pr(X=\omega) \right] \cdot \left[\sum_z \Pr(Y=z) \cdot z \right]$$

$$= \left[\sum_{\omega} \omega \cdot \Pr(X=\omega) \right] \cdot \left[\sum_z \Pr(Y=z) \cdot z \right]$$

$$= E[X] \cdot E[Y]$$

$E[XY] = E[X] \cdot E[Y] \Rightarrow X, Y$ are "uncorrelated".

Independence $\Rightarrow$ Uncorrelated. Uncorrelated $\nRightarrow$ Independence.

Markov's ^{least exciting} Inequality

For a non-negative random

variable X , $\Pr[X \geq a] \leq \mathbb{E}[X] / a$

Equivalent: $\Pr[X \geq c \cdot \mathbb{E}[X]] \leq 1/c$. Think $c = 2, 10, \dots$

Proof: $\mathbb{E}[X] = \sum_{\omega} \Pr[X = \omega] \cdot \omega$

$$= \underbrace{\sum_{\omega < a} \Pr[X = \omega] \cdot \omega}_{\geq 0} + \sum_{\omega \geq a} \Pr[X = \omega] \cdot \omega$$

$$\geq \sum_{\omega \geq a} \Pr[X = \omega] \cdot a$$

$$= a \cdot \sum_{\omega \geq a} \Pr[X = \omega]$$

$$= a \cdot \Pr[X \geq a]$$

$$\mathbb{E}[X] \geq a \cdot \Pr[X \geq a]$$

Is Markov's Inequality Tight?

Can you prove something better?

In general, Markov's is tight.

$X = 0$ with probability $1 - \frac{t}{a}$

$= a$ with probability t/a .

$$\mathbb{E}[X] = 0 \cdot \left(1 - \frac{t}{a}\right) + a \cdot \left(\frac{t}{a}\right) = t.$$

$$\mathbb{P}[X \geq a] = t/a = \mathbb{E}[X]/a.$$

Hashing (+ Load Balancing)

- Work horse of the modern web

- good probability review!

(key, value) store

3 operations :

- insert (k_1, v_1)
- insert (k_2, v_2)
- ⋮
- insert (k_m, v_m)

delete (k_{10}, v_{10})

⋮

query (k_j) $\rightarrow v_j$
query (k_{10}) $\rightarrow$ empty

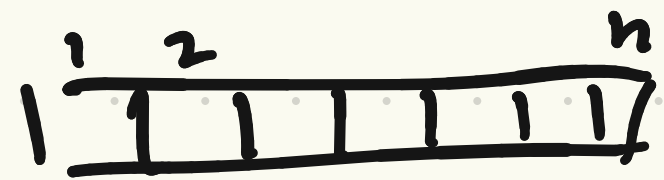
↓
Want

1) Fast queries:
 $O(1)$ time

2) Small space:
 $O(m)$ space

Hashing

- Build table T of size n
- Choose random function



$$h: U \rightarrow \{1, \dots, n\}$$

↓
universe of possible
keys

$$k_1, \dots, k_m \in U$$

<u>U</u>	<u>h</u>
u_1	$h(u_1) = 10$
u_2	$h(u_2) = 4$
u_3	$h(u_3) = m$
$\vdots$	$\vdots$
$u_{1000000}$	$h(u_{1000000}) = 4$

h drawn uniform from $\mathcal{H}$

all possible mappings from
 $U \rightarrow \{1, \dots, n\}$

"hash family"

$$(k \cdot r_1 + r_2) \pmod{m}$$

Is this possible
in practice?

3 Issues!

Hashing

- for $\text{insert}(k, v)$, store v at $T_h(k)$
- for $\text{delete}(k, v)$, remove v from $T_h(k)$
- for $\text{query}(k)$, look at $T_h(k)$

$m \ll |U|$ so could have $h(k) = h(j)$ for $j \neq k$.

"hash collision"

Store $v(k), v(j)$ in linked list at $T_h(k) = T_h(j)$

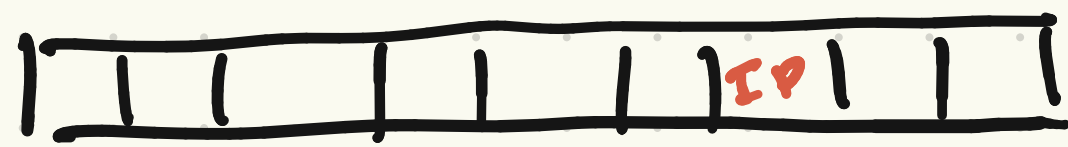
Goal: Hash collisions should be rare!

Lookup time = $O(1)$ if no collision,

otherwise $O(\text{length of linked list})$

Hashing Applications

- URL / DNS resolution. What IP address to visit to see `www.nyu.edu`?

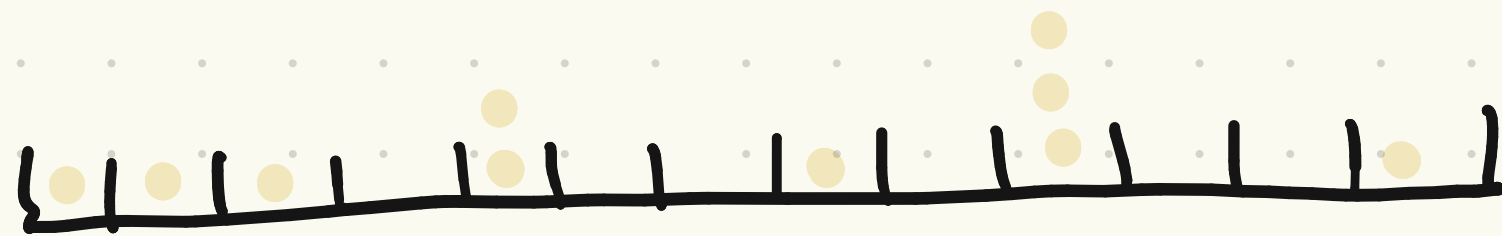


$\uparrow$
 $T_h(\text{"www.nyu.edu"})$

- Web content delivery (distributed hash table)
- Amazon DynamoDB, MongoDB, Cassandra, Google Datastore
"no-sql" data bases
- Google directions: send "Boston to NYC" to $A_h(\text{"Boston to NYC"})$
 $\downarrow$
servers $A_1, \dots, A_8$

Goal: Hash collisions should be rare!

How many collisions in expectation when inserting m items into table of size n ?



of collisions = random variable, with randomness coming from choice of $h \in \mathcal{H}$.
 C take $k_1, \dots, k_m$ as fixed.

$$C = \frac{1}{2} \sum_{i=1, \dots, m} \sum_{j \neq i} \mathbb{1}[h(k_i) = h(k_j)]$$

$$\begin{aligned} \mathbb{1}[\text{true}] &= 1 \\ \mathbb{1}[\text{false}] &= 0 \end{aligned}$$

↙
"indicator function"

How many collisions in expectation when inserting m items into table of size n?

$$\mathbb{E}[C] = \mathbb{E}\left[\frac{1}{2} \sum_i \sum_{j \neq i} \mathbb{1}[h(k_i) = h(k_j)]\right]$$

$$= \frac{1}{2} \sum_i \sum_{j \neq i} \underbrace{\mathbb{E}[\mathbb{1}[h(k_i) = h(k_j)]]}_{= 1/n}$$

$$= \frac{1}{2} \sum_i \sum_{j \neq i} 1/n = \boxed{\frac{m \cdot m - 1}{2n}}$$

Result 1 Collision free hash table with $O(m^2)$ space. ↗ $O(1)$ time lookups.

Set $n = 5m^2$. $E[C] = \frac{m(m-1)}{2n} \leq \frac{1}{10}$. $Pr[C \geq 1] \leq 1/10$.
Markov's inequality

Could keep retrying until achieve collision free hash.

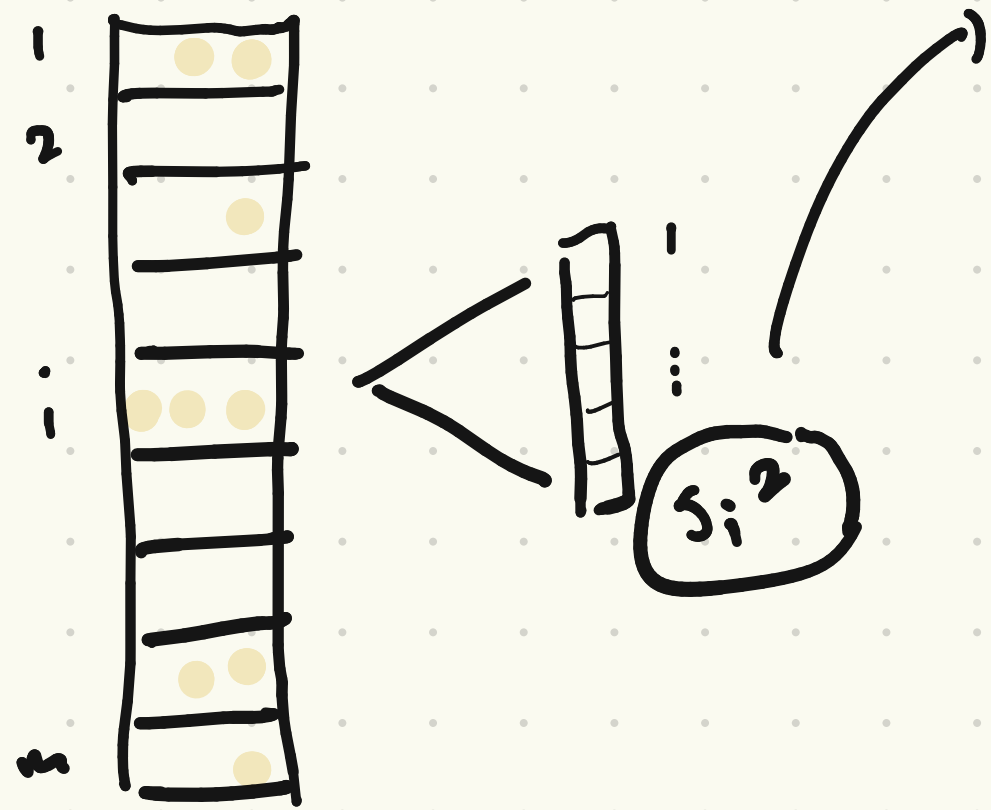
Trial 1:	1	2	3	4	...	6
Failure Probability:	$1/10$	$1/100$	$1/1000$	$1/10000$		$\leq$ chance getting struck by lightning.

$O(m^2)$ is a lot of space overhead for m items.

"Birthday Paradox"

Result 2 Collision free hash table with $O(m)$ space.

Key Idea: 2 Layer Hashing



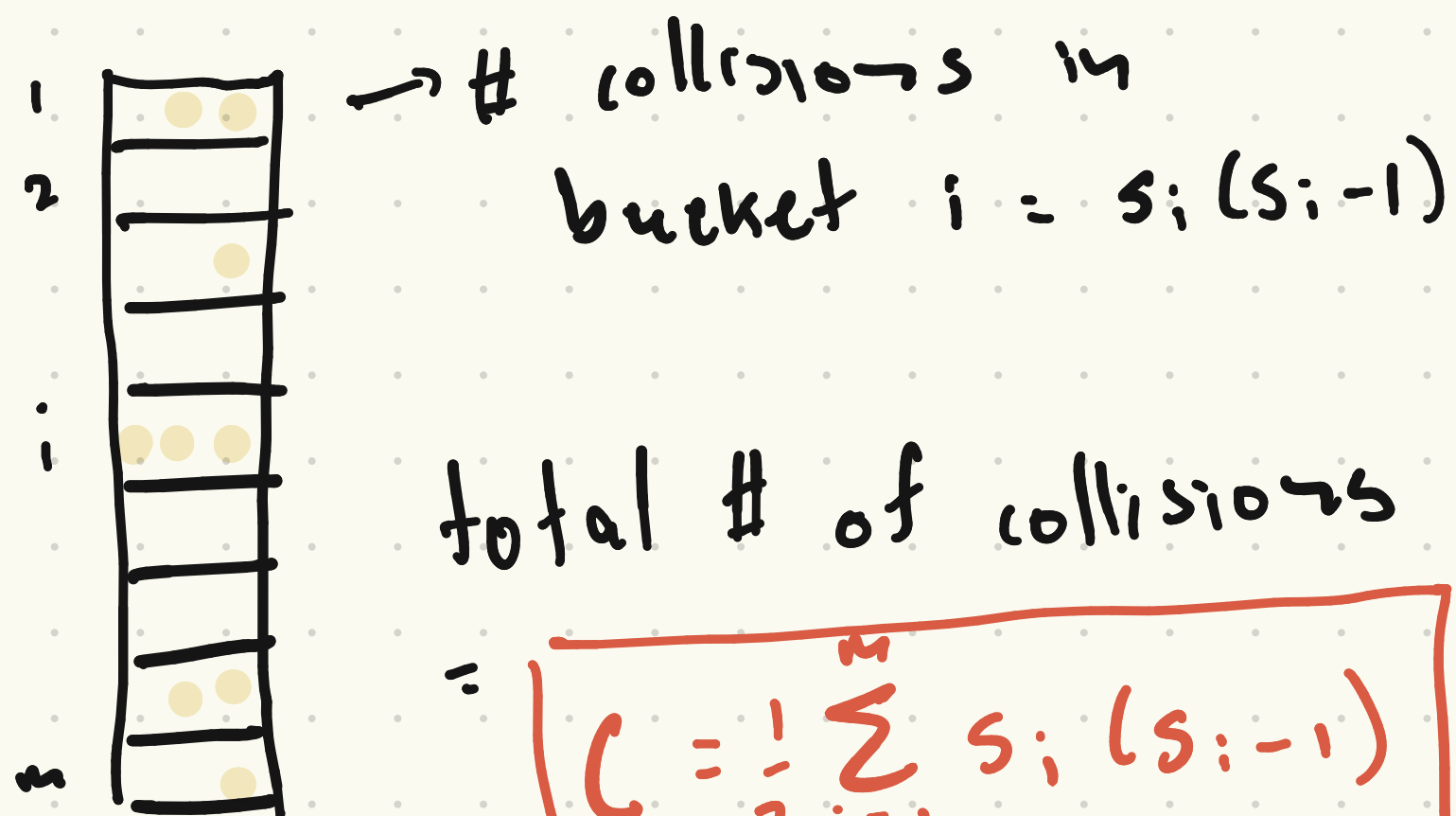
2nd level is collision free $\rightarrow$ each lookup takes just 2 hash function evaluations.

$$\text{Total Space} = m + \sum_{i=1}^m s_i^2$$

$$\mathbb{E} \left[m + \sum_{i=1}^m s_i^2 \right] = m + \underbrace{\mathbb{E} \sum_{i=1}^m s_i^2}_{?}$$

items in bin $i = s_i$

$$\mathbb{E}[\text{total space}] = m + \mathbb{E}\left[\sum_{i=1}^m s_i^2\right]$$



total # of collisions

$$C = \frac{1}{2} \sum_{i=1}^m s_i(s_i - 1)$$

→ almost s_i^2 !

Almost instant lookups
with only $3 \times$ space
overhead!

→ Markov's
inequality is
actually quite
powerful!

$$\mathbb{E} \sum_{i=1}^m s_i^2 = \mathbb{E} \left[\sum_{i=1}^m s_i(s_i - 1) + \sum_{i=1}^m s_i \right]$$

$$= 2 \mathbb{E}[C] + m < 2m$$

with m buckets, m items
- $\frac{m(m-1)}{2m} \leq \frac{1}{2}$

$$\mathbb{E}[\text{total space}] = 3m$$

- Google directions: send "boston to NYC" to $A_h(\text{"boston to NYC"})$
 $A_1, A_2, \dots, A_n$

why?

- Google directions: send "Boston to NYC" to $A_h(\text{"Boston to NYC"})$
 $A_1, A_2, \dots, A_n$

Now we care about the maximum # of elements per bin.

"Load Balancing" $\max[s_1, \dots, s_n]$

Suppose we hash n items to n servers. "Balls into Bins"
 $\rightarrow b_1, \dots, b_n$

What does Markov's give?

$$\mathbb{E}[s_i] = \mathbb{E}\left[\sum_{j=1}^n \mathbb{1}[h(b_j) = i]\right] = \sum_{j=1}^n \frac{1}{n} = 1.$$

$$\boxed{\Pr[s_i \geq 10] \leq 1/10}$$

1 in 10 servers could be overloaded! No bound on max.

Goal: For any i , $\Pr[s_i \geq B] \leq \frac{1}{10n}$.

Corollary: With probability $9/10$ all $s_i \leq B$.

Proof: Union bound: Given random events A, B

$$\Pr[A \text{ or } B \text{ occur}] \leq \Pr[A] + \Pr[B]$$

$$\begin{aligned} &= \Pr[A + \bar{B}] + \Pr[\bar{A} + B] - \Pr[A + B] \\ &= \Pr[A + \bar{B}] + \Pr[B] \leq \Pr[A] + \Pr[B] \end{aligned}$$

$$\Pr[s_1 \geq B \text{ or } s_2 \geq B \text{ or } \dots s_n \geq B] \leq \frac{1}{10n} + \frac{1}{10n} + \dots + \frac{1}{10n} = \frac{1}{10}.$$

Goal: For any i , $\Pr[s_i \geq \beta] \leq \frac{1}{10n}$.

Markov's: $\beta = 10n$. $\rightarrow$ Vacuous bound! Only n items...

Tighter bounds on deviation by considering more information about a random variable.

Variance: For r.v. X , $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$

Equivalent: $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

Proof: $\mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2]$

$$\text{Var}(tX) = t \text{Var}(X)$$

Linearity of
Variance:

$$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] \text{ if } X, Y \text{ independent}$$

$$\text{Var}[X_1 + X_2 + \dots + X_m] = \text{Var}[X_1] + \text{Var}[X_2] + \dots + \text{Var}[X_m]$$

if X_i, X_j independent for all $i, j \in 1, \dots, m$.

"pairwise independence"

"mutual independence" is stronger. why?

Chebyshev Inequality: Let X be a r.v. with $\text{var}(X) = \sigma^2$.

$$\Pr[|X - \mathbb{E}X| \geq k\sigma] \leq 1/k^2$$

↑
upper and lower bound

—no assumption that $X \geq 0$.

Proof. Consider the random variable $Y = (X - \mathbb{E}X)^2$.
 Y is non negative. By Markov inequality,

$$\Pr[Y \geq k^2 \mathbb{E}[Y]] \leq 1/k^2$$

$$\mathbb{E}[Y] = \mathbb{E}[X - \mathbb{E}X]^2 = \text{var}(X) = \sigma^2.$$

$$\Pr[(X - \mathbb{E}X)^2 \geq k^2 \sigma^2] \leq 1/k^2$$

Goal: For any i , $\Pr[S_i \geq \beta] \leq \frac{1}{10n}$

$\downarrow$
 $\#$ of balls to bin i

$$S_i = \sum_{j=1}^n \mathbb{I}[h(b_j) = i] \quad \text{Var}[S_i] = \sum_{j=1}^n \text{Var} \underbrace{\mathbb{I}[h(b_j) = i]}_{X_j}$$

$$\text{Var}(X_j) = \mathbb{E}[X_j^2] - \mathbb{E}[X_j]^2 = \boxed{\frac{1}{n} - \frac{1}{n^2}}$$

$$X_j = \begin{cases} 1 & \text{w/ prob. } 1/n \\ 0 & \text{otherwise} \end{cases}$$

$$X_j^2 = \begin{cases} 1 & \text{w/ prob } 1/n \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X_j] = 1/n$$

$$\mathbb{E}[X_j^2] = 1/n$$

$$\text{Var}[S_i] = n \left(\frac{1}{n} - \frac{1}{n^2} \right) = 1 - \frac{1}{n} \leq 1.$$

Goal: For any i , $\Pr[s_i \geq \beta] \leq \frac{1}{10n}$.

(Chebyshev's: $\Pr[|x - \mathbb{E}x| \geq k\sigma] \leq 1/k^2$)

$$\text{Var}[s_i] = 1 - 1/n.$$

$$\mathbb{E}[s_i] = 1$$

Improvable to $O(\log n)$!

Set $k = \sqrt{10n}$.

$$\Pr[|s_i - 1| \geq \sqrt{10n} \cdot \sqrt{1 - 1/n}] \leq \frac{1}{10n}$$

$$\Pr[s_i - 1 \geq \sqrt{10n} \cdot 1] \leq 1/10$$

With probability $9/10$
all bins have $\leq O(\sqrt{n})$ balls.

$$\Pr[s_i \geq O(\sqrt{n})] \leq 1/10n$$

$n = 1,000,000$

max load ≈ 1000 .

In class exercise



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New York University Tandon School of Engineering
Computer Science and Engineering

CS-GY 9223I: Lecture 1 Coursework

Problem 1: Hash collisions are useful?

Your company is considering paying for a cloud service that provides CAPTCHA-like visual puzzles for verifying that users are human. The company providing the service claims to have a larger database of unique puzzles than any competitors, but you don't trust the salesperson.

- (a) The company has provided you with an API end-point which returns puzzles uniformly and independently at random from their database. Using this endpoint, describe a simple randomized estimator for the number of puzzles in the database, n .
- (b) The company claims their database has 1,000,000 unique CAPTCHAs in it. Using your estimator, roughly how many queries do you need to verify their claim with good probability (e.g. 9/10)? You should need far less than 1,000,000 queries!
- (c) More generally, how many samples are required to estimate the true number of CAPTCHAs, n , in the database up to additive error $\pm \epsilon n$, with good probability?

(a) $X = \#$ of repeated pairs in m draws

$c_1, \dots, c_m \in \{1, \dots, n\}$

$$X = \frac{1}{2} \sum_i \sum_{j \neq i} \mathbb{1}[c_i = c_j]$$

$$\mathbb{E}[X] = \frac{m(m-1)}{2n}$$

Estimate $n = \frac{m(m-1)}{2X}$

If X is close to $\mathbb{E}[X]$, $\frac{m(m-1)}{2X} \approx \frac{m(m-1)}{2\mathbb{E}[X]} = n$.

$$\text{Var}(X) = \frac{1}{2} \sum_i \sum_{j \neq i} \text{Var}[\mathbb{1}[c_i = c_j]] \leq \frac{m(m-1)}{2n}$$

If $m = O\left(\frac{\sqrt{n}}{\epsilon}\right)$, $\text{Var}(X) \leq \frac{1}{n}$. $\mathbb{E}[X] \approx \frac{1}{4\epsilon^2}$

$$\Pr\left[|X - \mathbb{E}[X]| \geq \frac{1}{4}\right] \leq \frac{1}{10}, \quad \Pr\left[|X - \mathbb{E}[X]| \geq \epsilon \mathbb{E}[X]\right] \leq \frac{1}{10}$$

$$X \in [(1-\epsilon)\mathbb{E}[X], (1+\epsilon)\mathbb{E}[X]]$$

Useful identities: $\frac{1}{1+\epsilon} \geq 1-\epsilon$

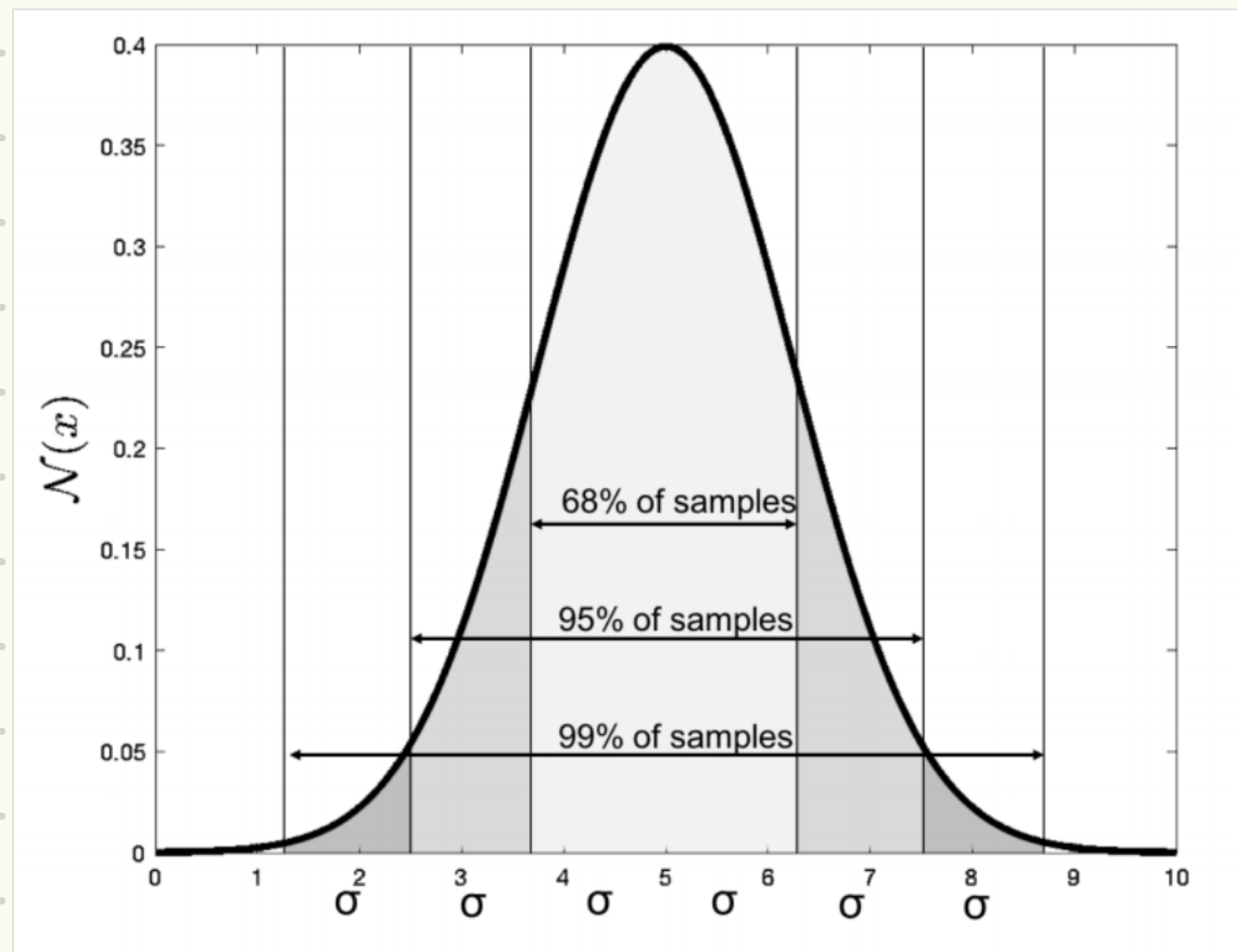
$$\frac{1}{1-\epsilon} \leq 1+2\epsilon \quad \text{for } \epsilon \leq 1/2.$$

$$\frac{1}{1+\epsilon} X \leq Y \leq \frac{1}{1-\epsilon} X \quad \rightarrow \quad (1-\epsilon) X \leq Y \leq (1+2\epsilon) X$$

Chernoff Bounds

Motivating question: How tight is Chebyshev's inequality?

Truth



$$\Pr[|X - \mathbb{E}X| \geq 1\sigma] = 32\%$$

$$\Pr[|X - \mathbb{E}X| \geq 2\sigma] = 5\%$$

$$\Pr[|X - \mathbb{E}X| \geq 3\sigma] = 1\%$$

$$\Pr[|X - \mathbb{E}X| \geq 4\sigma] = .01\%$$

Chebyshev

$$\Pr[|X - \mathbb{E}X| \geq 1\sigma] \leq 100\%$$

$$\Pr[|X - \mathbb{E}X| \geq 2\sigma] \leq 25\%$$

$$\Pr[|X - \mathbb{E}X| \geq 3\sigma] \leq 11\%$$

$$\Pr[|X - \mathbb{E}X| \geq 4\sigma] \leq 6\%$$

Normal / Gaussian random variable
 $X \sim N(0, \sigma^2)$

Key Takeaway: Gaussian r.v.'s concentrate much more tightly around their expectations than variance alone can predict.

Probability density of Gaussian:

$$p(x) \sim e^{-x^2/2\sigma^2}$$

} exponential decay as x grows

Lemma: For $x \sim N(\mathbb{E}x, \sigma^2)$,

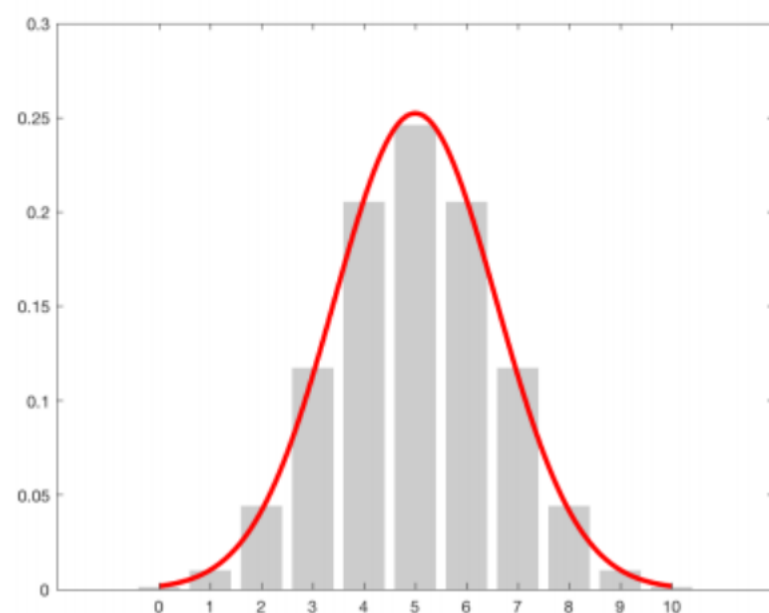
$$\Pr(|x - \mathbb{E}[x]| > k\sigma) \leq O(e^{-k^2/2})$$

vs. $O(1/k^2)$ for general r.v.'s

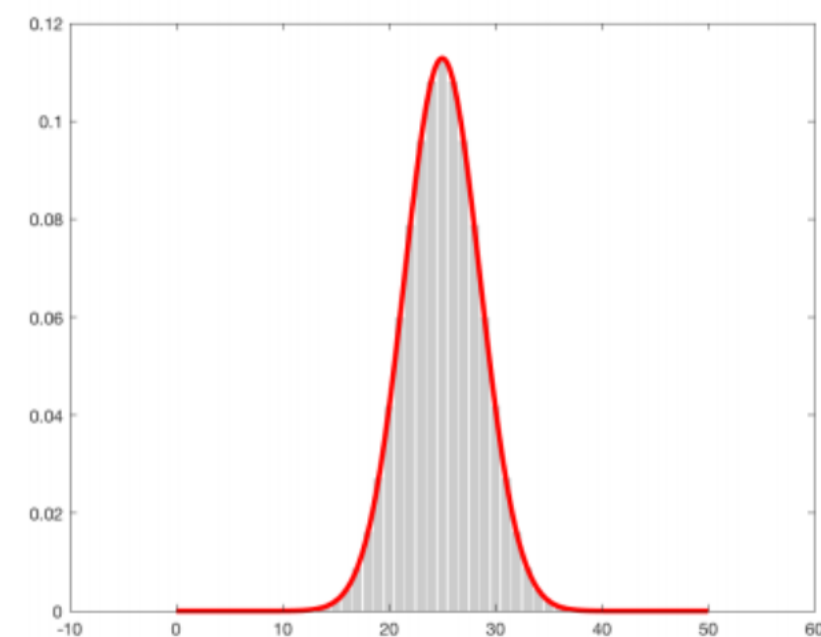
Why does this matter for algorithm design?

Central Limit Theorem (informal):

Any sum of n independent random variables $X_1, \dots, X_n$, each with bounded mean and variance, converges to a Gaussian random variable.



(a) Distribution of # of heads after 10 coin flips, compared to a Gaussian.



(b) Distribution of # of heads after 50 coin flips, compared to a Gaussian.

Example: Repeated tosses of unbiased random coin.

Back of the envelope calculation:

$$X_i = \begin{cases} 1 & \text{w/ prob. } 1/2 \\ 0 & \text{w/ prob. } 1/2 \end{cases} \quad S = \sum_{i=1}^n X_i$$

$$i = 1, \dots, n$$

$$\sigma^2 = \text{Var}(S) = \sum_{i=1}^n \text{Var}(X_i) = n \cdot \mathbb{E}[|X - \mathbb{E}X|^2] = n/4$$

$$S \approx N(0, n/4)$$

$$\Pr(|\bar{x} - \mu| \geq \underset{\downarrow \sqrt{n}}{k \epsilon}) \leq e^{-k^2/2}$$

To get error ϵn , need $k = \epsilon \sqrt{n}$. If $n = O(1/\epsilon^2)$, holds with prob. $e^{-O(1)} \leq 1/10$. If $n = O(\log^4 n / \epsilon^2)$ holds with prob. $e^{-O(\log^4 n)} \leq 1/n^{10}$

Pay very little for much higher confidence bound!

In general, require $O(\log^{(1/\delta)}(1/\delta) / \epsilon^2)$ samples to get error ϵ with $1-\delta$ probability. (Chebyshev's would have predicted $1/\epsilon^2 \delta^2$).

Formal Theorem: "Bernstein Inequality"

Let $X_1, X_2, \dots, X_n$ be independent random variables and each $X_i \in [-1, 1]$. Let $\mu_i = \mathbb{E}[X_i]$ and $\sigma_i^2 = \text{var}[X_i]$.

Then $S = \sum_{i=1}^n X_i$ satisfies:

$$\Pr[|X - \mu| \geq k\sigma] \leq 2e^{-k^2/4}$$

where $\mu = \sum \mu_i$ and $\sigma^2 = \sum \sigma_i^2$, under the condition that $k \leq \frac{1}{2}\sigma$.

Similar bounds with different names:

(Chernoff bound, Hoeffding bound).

Independent : "mutually independent" :

If X, Y, Z are R.V.s taking values in S_x, S_y, S_z

then for all $a \in S_x, b \in S_y, c \in S_z$,

$$\Pr(X=a, Y=b, Z=c) = \Pr(X=a) \Pr(Y=b) \Pr(Z=c)$$

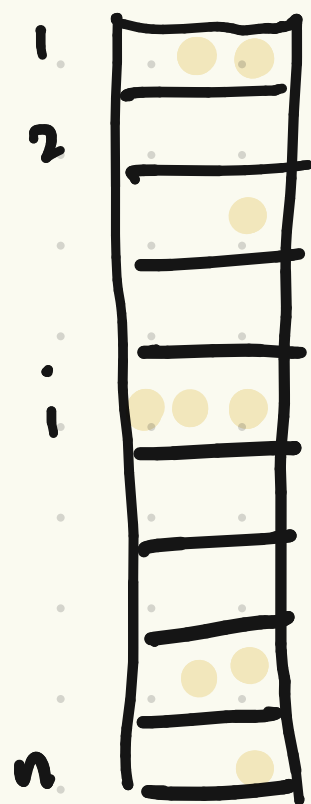
Stronger than pairwise independence.

Chernoff Bound Let $X_1, X_2, \dots, X_n$ be independent
0/1 random variables and let $p_i = \mathbb{E}[X_i]$, where $0 \leq p_i \leq 1$.
Then $S = \sum_{i=1}^n X_i$ which has mean $\mu = \sum_{i=1}^n p_i$ satisfies:

$$\Pr[S \geq (1+\epsilon)\mu] \leq e^{-\epsilon^2 \mu / 3 + 3\epsilon}$$

Application to Load Balancing

n balls, n bins, read only assigned



$S_i = \#$ of balls in bin i .

$$S_i = \sum_{j=1}^n \mathbb{1}[h(b_j)=i]$$

X_j

$$\mathbb{E}[X_j] = 1/n \quad \text{Var}[X_j] = \frac{1}{n} - \frac{1}{n^2}$$

$$\mu = \mathbb{E}[S_i] = 1$$

$$\sigma^2 = \text{Var}[S_i] = n \left(\frac{1}{n^2} - \frac{1}{n} \right) \leq 1$$

set $\epsilon = 6 \log n$.

$$\Pr[S_i \geq 1 \cdot (1 + 6 \log n)] \leq e^{-\epsilon^2 / 3 + 3\epsilon} \leq e^{-\epsilon/3} = e^{-2 \log n} = \frac{1}{n^2}$$

By a union bound, only a $n \cdot \frac{1}{n^2} = \frac{1}{n}$ chance that any bin has load $\geq O(\log n)$.